

Daniel Whitehead

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EDUCATION

Duke University - The Nicholas School of the Environment

Durham, North Carolina

Master of Environmental Management

May 2026

Concentrations: Energy and Environment, Environmental Analytics and Modeling

Relevant Coursework: Energy System Modeling, Energy Finance, Energy Economics, Markets for Electric Power, Machine Learning

Xavier University - College of Arts and Sciences

Cincinnati, Ohio

Bachelor of Science (3+2 Acceleration)

May 2025

Major: Environmental Science | Minor: Peace and Justice Studies | Cum Laude

Relevant Coursework: Natural Resource Economics, Professional Communication in Science, Climate Change

WORK EXPERIENCE

PSG Energy Group

Remote

Energy Analyst, Portfolio and Project Development

June 2025 - Present

- Used Python and Excel to monitor operational performance of a 30+ MW, 80-site C&I solar portfolio, assess compliance with production-guarantee requirements (AlsoEnergy), and build automated data pipelines to Power BI dashboards.
- Managed RFP's, established timelines for procurement and utility service, and analyzed renewable project economics, integrating energy market data, utility tariffs, and rate structures to develop proformas which quantified operational risks.
- Built financial models and Renewable Energy Certificate strategy models guiding >\$1M procurement.

The World Bank Group

Remote

Graduate Research Analyst

May 2025 - May 2026

- Developed a performance-based incentive model to improve grid reliability in African power markets, analyzing demand volatility, grid constraints, generation mix, fuel costs, outages, and regulatory impacts on system reliability for 29 nations.
- Collaborated with the World Bank Chief Economist (Africa) to evaluate debt-serviceability and tax-incentive models, utilizing Excel, Python, and SQL for data analysis; communicated findings to technical and non-technical audiences.

Xavier University

Cincinnati, Ohio

Student Research Analyst - Experimental Biology Lab

August 2023 - December 2023

- Designed and executed research investigating macroinvertebrates within local stream ecosystems to assess low-head dam impacts on water quality and environmental health; collected and conducted statistical analysis of 514 macroinvertebrates.

Kurt Vonnegut Museum and Library

Indianapolis, Indiana

Museum and Merchandise Manager

August 2018 - August 2024

- Directed on-site operations during high-volume summer shifts and developed an Excel-based staffing and inventory model that improved operational efficiency by 30%.

ACADEMIC PROJECTS (Duke University)

CAISO and ERCOT Energy Load Forecasting

Durham, North Carolina

- Using Python and R, developed neural network and machine-learning forecasting models (using Scikit-learn and XGBoost) to predict load volatility and peak-risk events using large historical datasets (weather, demand, generation).

Power Market Analysis: Stress-testing ISO New England Under Seasonal Volatility

Durham, North Carolina

- Analyzed ISO New England grid resilience in Python using EPA eGRID2023 and EIA fuel price data to evaluate power generation assets and system capacities: quantified the risk of extreme weather and fuel supply chain vulnerabilities.

Regional Power Analysis: 95% Decarbonization & Investment Roadmap

Durham, North Carolina

- Structured a 95% decarbonization roadmap for a regional system operator, optimizing long-term resource adequacy and grid reliability through sensitivity-driven risk modeling of a diversified 10-asset energy portfolio (using Python, Excel, SQL).

Renewable Energy Project Finance Modeling (Wind & BESS)

Durham, North Carolina

- Engineered dynamic wind and BESS financial models to calculate IRR, executing multi-variable sensitivity analyses across leverage structures (40%-80%), CAPEX, and O&M escalators to stress-test asset valuations.

SKILLS AND CERTIFICATIONS

Bloomberg Market Concepts Certification, Data Analysis and Visualization, Python/Excel/Power BI/SQL, Energy Load Forecasting